Testing a satellite water balance for groundwater recharge in tropical basement catchments

Upper Ouémé, Benin, and Upper Ogun, Nigeria, 2018–2024

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Catchments	Upper Ouémé at Bétérou, Benin, 10,102 km² (test); Upper Ogun, Nigeria, 9,534 km² (application)
Period	2018–2024; rainfall baseline 1991–2020; river-flow record 1971–2010
Data	CHIRPS v2.0 and v3.0, TAMSAT v3.1, FAO WaPOR v3, HydroBASINS, published Bétérou flows
Keywords	groundwater recharge, water balance, crystalline basement, satellite rainfall, evapotranspiration, uncertainty
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Report version of the web publication at oloyededavid.github.io/Oueme-Ogun-Groundwater-Recharge. Code, data and tests: github.com/oloyededavid/Oueme-Ogun-Groundwater-Recharge.

Abstract

Groundwater in the weathered crystalline basement of Benin and south-west Nigeria supplies much of the rural water, but recharge to these aquifers is poorly quantified and few wells are monitored. Satellite rainfall and evapotranspiration cover every catchment, which makes a water balance an attractive way to estimate recharge where wells are absent. This study tests that approach in the Upper Ouémé at Bétérou (Benin, 10,102 km²), where historical river flow is published, and applies it unchanged to the Upper Ogun (Nigeria, 9,534 km²) for 2018–2024. Rainfall comes from CHIRPS v2.0, CHIRPS v3.0 and TAMSAT v3.1, and actual evapotranspiration from FAO WaPOR v3. No discharge is available for 2018–2024, so river flow was estimated from the runoff ratio observed at Bétérou in 1988–2010. With that ratio, the Upper Ouémé balance leaves a residual of 122–169 mm per year (9.8–13.0% of rainfall), positive in every year and with every rainfall product. The residual cannot be water stored in the aquifer: holding it would need a water-table rise of at least 57 m. It is, however, conditional on the transferred runoff ratio. A ratio of 0.24–0.27, inside the single-year range reported for a sub-basin of the same catchment, closes the balance. The residual is as large as recharge estimated from groundwater levels in Benin basement, about 4% and 5–24% of rainfall. With current satellite products, and without contemporary discharge and well records, a catchment water balance cannot resolve recharge here.

Highlights

- With the 1988–2010 runoff ratio, the 2018–2024 balance at Bétérou leaves 122–169 mm per year, positive in all seven years.
- The residual is not stored groundwater: storing it would need a water-table rise of at least 57 m.
- A runoff ratio of 0.24–0.27 instead of 0.142 would close the balance, so the result depends on river flow that was not measured.
- At 9.8–13.0% of rainfall, the residual is as large as published recharge (4% and 5–24%).
- WaPOR v3's precipitation layer equals CHIRPS v3.0 to within 0.25 mm per year, so the two are one estimate, not two.

1 Introduction

Across much of tropical Africa, groundwater is held in weathered crystalline rock. A synthesis of 134 ground-based estimates found these aquifers to combine low storage with high, regular recharge, and found long-term rainfall to be the only factor related to long-term recharge at continental scale (MacDonald et al., 2021). In Benin, groundwater-level records give recharge of about 4% of rainfall in a weathered-crystalline aquifer over 19–25 years (Kotchoni et al., 2019) and 5–24% over two years in the Donga sub-basin of the Upper Ouémé (Kamagaté et al., 2007). Such records exist at few sites.

Recharge methods differ in the time and space scales they represent, and estimates from more than one method are needed before any one is trusted (Scanlon et al., 2002). The water-table fluctuation method needs well records and a specific yield (Healy & Cook, 2002). Where neither exists, a catchment water balance is an alternative: rainfall minus evapotranspiration minus river flow leaves the water that recharged the aquifer or changed storage. Satellite products now provide the first two terms everywhere.

Each term carries an error that may be as large as the recharge it is meant to reveal. WaPOR v2 evapotranspiration differed from flux-tower measurements across Africa by 29–31% (Blatchford et al., 2020), rainfall products over Nigeria had biases of 20–26% (Ogbu et al., 2020), and river flow is rarely measured in the same years as the satellite record. Whether a satellite balance is precise enough for recharge therefore has to be tested where an independent check exists, before it is applied where none does.

This study makes that test in the Upper Ouémé, where river flow at Bétérou has been published, and asks three questions:

RQ1. Does a satellite water balance close against the river flow observed at Bétérou?

RQ2. What does the residual represent, and how sensitive is it to the rainfall product, the evapotranspiration estimate and the runoff assumption?

RQ3. What can the same method, applied unchanged, say about the ungauged Upper Ogun?

2 Study area

The **Upper Ouémé** is the part of the Ouémé basin upstream of the Bétérou gauge in central Benin. The northern Ouémé lies on the Dahomeyan crystalline basement and has a Sudanian climate with a single rainy season at Bétérou (Kodja et al., 2020). In such terrain, groundwater is held mainly in a weathered layer (saprolite) above fissured rock, mostly within the upper 100 m (Lachassagne et al., 2021). In the Donga sub-basin (586 km^2), inside the catchment, recharge is by direct infiltration of rainfall, the weathered layer is up to a few tens of metres thick, and borehole abstraction is a small share of recharge (Kamagaté et al., 2007). Rainfall in 2018–2024 averaged 1,257 mm per year (CHIRPS v2.0).

The **Upper Ogun** is the Ogun basin in south-west Nigeria upstream of its confluence with the Oyan River. It averaged 1,311 mm of rainfall per year in 2018–2024. No published recharge estimate or hydrogeological study specific to this catchment was found. It lies north of the coastal sedimentary basin, and the weathering-profile description above is assumed, not shown, to apply.

Both catchments were traced upstream from a fixed outlet through the HydroBASINS level-9 network (Lehner & Grill, 2013). Their areas, computed on the WGS84 ellipsoid from the polygons analysed here, are 10,102 km^2 and 9,534 km^2. Every calculation uses these areas. HydroBASINS' own upstream-area attribute gives 10,143 and 9,575 km^2, and published areas at Bétérou are 10,070–10,076 km^2.

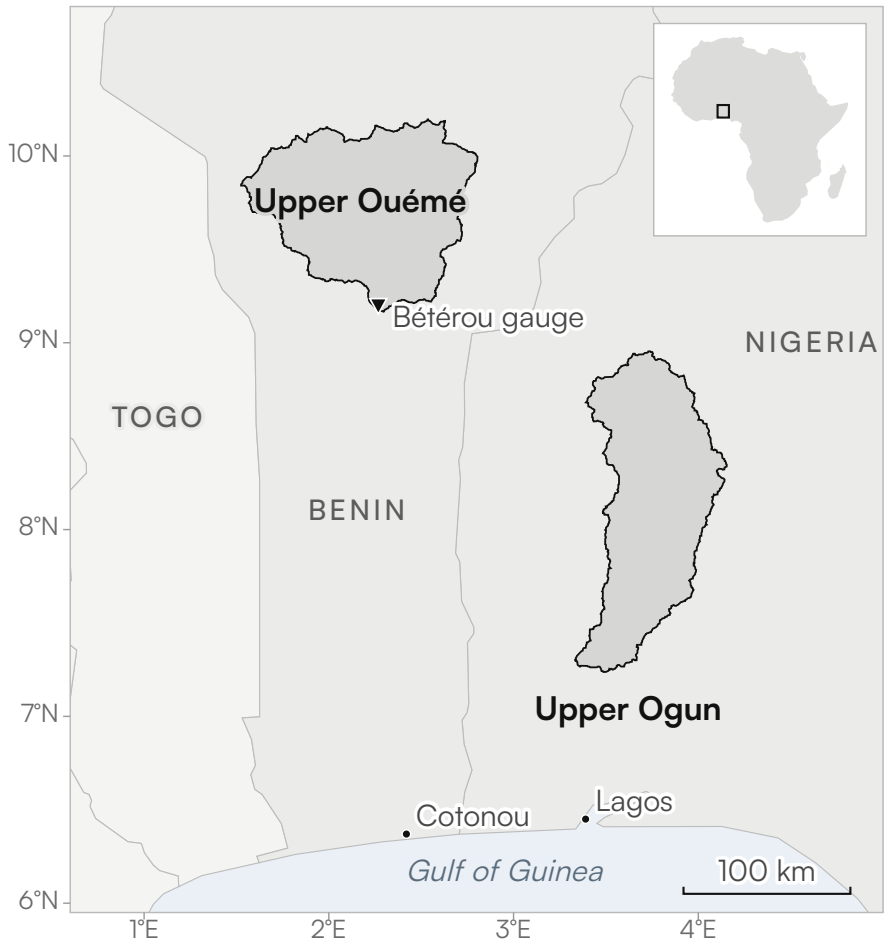


Figure 1. The Upper Ouémé and Upper Ogun catchments (HydroBASINS level 9) and the Bétérou river gauge. Inset: location in Africa. Country outlines from Natural Earth (1:50 m); latitude and longitude in degrees, WGS84.

3 Data and method

Table 1. Data used. All rainfall and evapotranspiration grids were summed to calendar years and averaged over each catchment.

Variable	Product	Resolution	Time step read	Years	Reference
Rainfall	CHIRPS v2.0	0.05°	annual	1981–2024	Funk et al., 2015
Rainfall	CHIRPS v3.0 (Africa)	0.05°	annual	1981–2024	Climate Hazards Center
Rainfall	TAMSAT v3.1, `rfe_filled`	0.0375°	monthly	1988–2010, 2018–2024	Maidment et al., 2017
Rainfall (check only)	WaPOR v3 L1-PCP	0.05°	annual	2018–2024	FAO
Evapotranspiration	WaPOR v3 L1-AETI (actual evapotranspiration and interception)	300 m	annual	2018–2024	FAO
River flow	Mean flow at Bétérou, with gauge-based catchment rainfall	Gauge	period means	1971–2010	Kodja et al., 2020
Catchments	HydroBASINS v1c, level 9	Vector	—	—	Lehner & Grill, 2013

The rainfall products are not independent. CHIRPS v2.0 and v3.0 share a method, WaPOR's precipitation layer is CHIRPS v3.0 (Section 4.1), and all three products are adjusted to rain gauges, of which there are few. Their spread is therefore a lower bound on rainfall uncertainty, not a measure of it. No gauge record for 2018–2024 was available, so agreement between products is reported, not error.

1

Catchments

HydroBASINS level 9 traced upstream from the Bétérou gauge and the Oyan confluence

2

Satellite terms

CHIRPS v2.0, v3.0 and TAMSAT rainfall; WaPOR v3 evapotranspiration; annual totals averaged over each catchment

3

River flow

Runoff ratio from published 1971–2010 flows; applied to each year's rainfall

4

Residual

$\varepsilon = P - ET - Q$ for each product and year

5

Tests

Storage, published recharge, runoff-ratio and single-term sensitivity, product agreement

Figure 2. Workflow. Steps 1–4 are scripts 01–03 of the repository; the tests in step 5 are in script 03.

3.1 Catchment averages

Annual totals were averaged over each catchment, weighting pixels by area and including every pixel whose centre lies inside the boundary. A catchment-year was accepted only if at least 95% of its area had valid data; every catchment-year did. WaPOR's 300 m evapotranspiration was averaged onto the 5 km rainfall grid for the maps. Each year was classed wet, normal or dry against the terciles of 1991–2020 CHIRPS rainfall.

3.2 Water balance

Over several years, changes in stored water should be small, so rainfall P, actual evapotranspiration ET and river flow Q should leave a residual ε close to zero (Eq. 2). Any difference also contains changes in storage, groundwater leaving the catchment underground, abstraction and errors in each term. No river flow was available for 2018–2024, so four kinds of quantity enter the balance, and they are kept apart here:

- **Observed.** Mean flow at Bétérou of 47.5 m³/s in 1971–1987 and 54.5 m³/s in 1988–2010, with catchment rainfall of 1,091 and 1,202 mm from 30 rain gauges (Kodja et al., 2020).
- **Derived.** Runoff ratios r of 0.136 and 0.142: each flow, as a depth over the catchment, divided by the gauge rainfall of the same period. The higher, r = 0.142, gives the smaller residual and is used throughout.
- **Estimated.** River flow for each year of 2018–2024, r times that year's satellite rainfall (Eq. 1).
- **Residual.** ε from Eq. 2. It is not called recharge.

$$Q = r\,P \tag{1}$$

$$\varepsilon = P - ET - Q \tag{2}$$

$$\Delta h = \Sigma\varepsilon \,/\, S_y \tag{3}$$

3.3 Tests of the residual

- **Runoff ratio.** The residual for ratios of 0.10–0.30, compared with the single-year ratios of 0.14–0.28 reported in the Donga sub-basin (Kamagaté et al., 2007); and the ratio recomputed from each product's own 1988–2010 rainfall instead of the gauge rainfall.
- **Storage.** The water-table rise Δh (Eq. 3) needed to store the 2018–2024 residual, for a specific yield S_y of 0.2–1.5% (Lachassagne et al., 2021), and the specific yield that would hold it in a rise of 10 m. This is a test of a hypothesis, not an estimate of storage change.
- **Scale.** Comparison with recharge from groundwater levels in Benin basement: about 4% of rainfall (Kotchoni et al., 2019) and 5–24% (Kamagaté et al., 2007).
- **Single terms.** The change in ET, P or r that sets the mean residual to zero, one term at a time. Lowering P also lowers Q (Eq. 1).
- **Product agreement.** Correlation and mean absolute difference between the annual rainfall of each pair of products (seven years, so indicative only).

4 Results

4.1 Rainfall and evapotranspiration

In 2018–2024 the Upper Ouémé received 1,243–1,298 mm of rainfall per year, depending on the product, and the Upper Ogun 1,296–1,492 mm (Figure 3). Both were wetter than 1991–2020, by 4.0% and 8.2% (CHIRPS v2.0). The Upper Ouémé had no dry year; the Upper Ogun had one, 2020, by CHIRPS v2.0 only. The study period does not

represent drought conditions.

The products agree on the mean better than on individual years. In the Upper Ouémé the correlation between their annual totals ranges from $r = -0.26$ to 0.63 across the three pairs, with mean absolute differences of 45–89 mm: CHIRPS v2.0 and TAMSAT do not even agree on which years were wetter. In the Upper Ogun the correlations are 0.74 to 0.89, but the differences larger, 66–196 mm. With seven years these correlations are indicative only. CHIRPS v3.0 is on average 41 mm wetter than v2.0 in the Upper Ouémé and 181 mm wetter in the Upper Ogun. WaPOR v3's precipitation layer, documented only as "CHIRPS", matches v3.0 to within 0.25 mm in every catchment-year.

WaPOR evapotranspiration was 945 mm per year in the Upper Ouémé (75% of rainfall) and 946 mm in the Upper Ogun (72%). It varied less from year to year than rainfall, with a coefficient of variation of 3.1% against 3.6–7.3% for rainfall in the Upper Ouémé. It is the largest term in the balance: an error of 10% in it would change the residual by 95 mm.

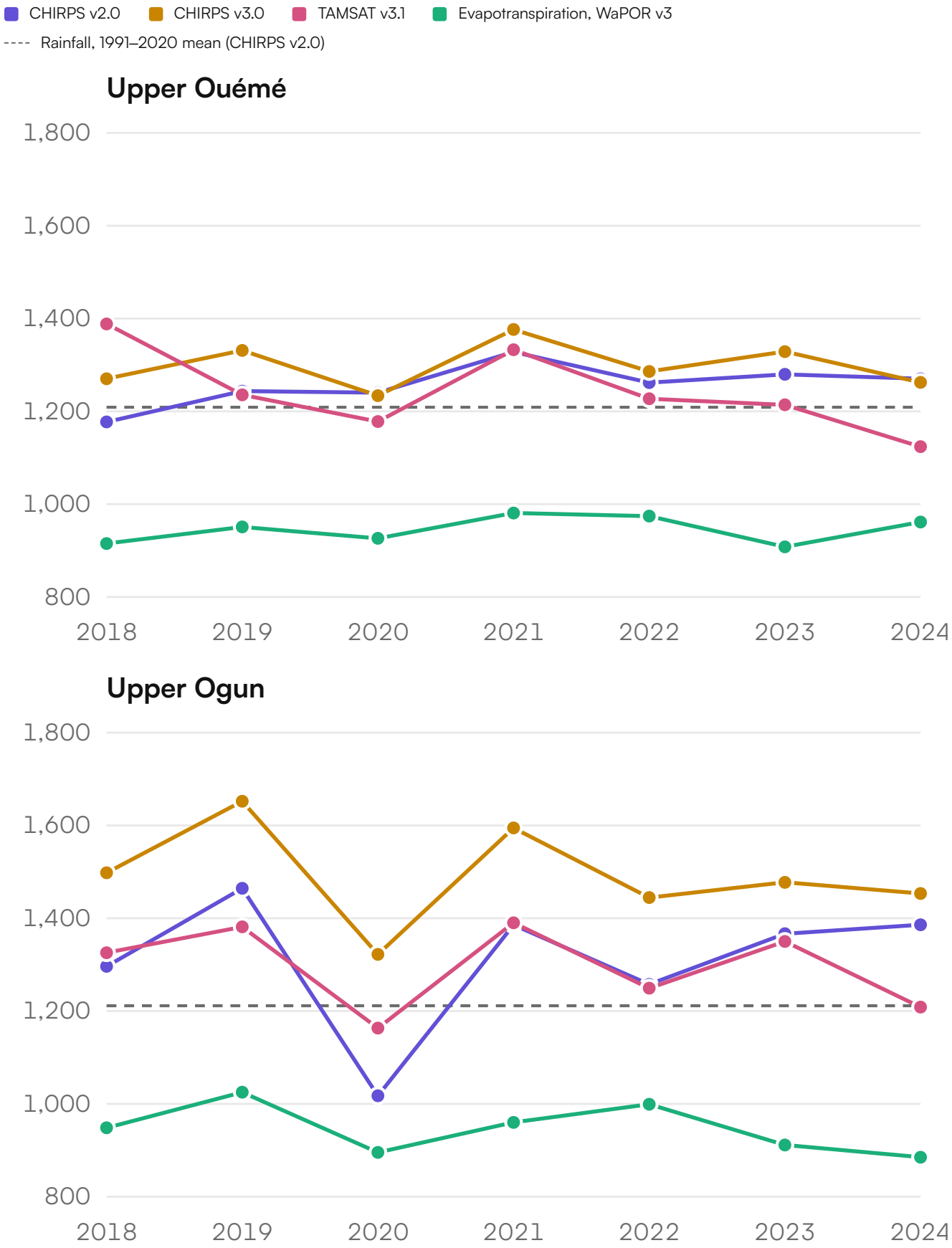


Figure 3. Annual catchment rainfall for three products and WaPOR v3 evapotranspiration, 2018–2024, mm per year. Dashed line: 1991–2020 mean rainfall, CHIRPS v2.0. WaPOR's precipitation layer is not drawn because it equals CHIRPS v3.0.

4.2 Water balance at Bétérou

With r = 0.142, the estimated river flow at Bétérou is 178 mm per year and the residual 122–169 mm per year, depending on the product (Table 2). It is positive in every year (Figure 4).

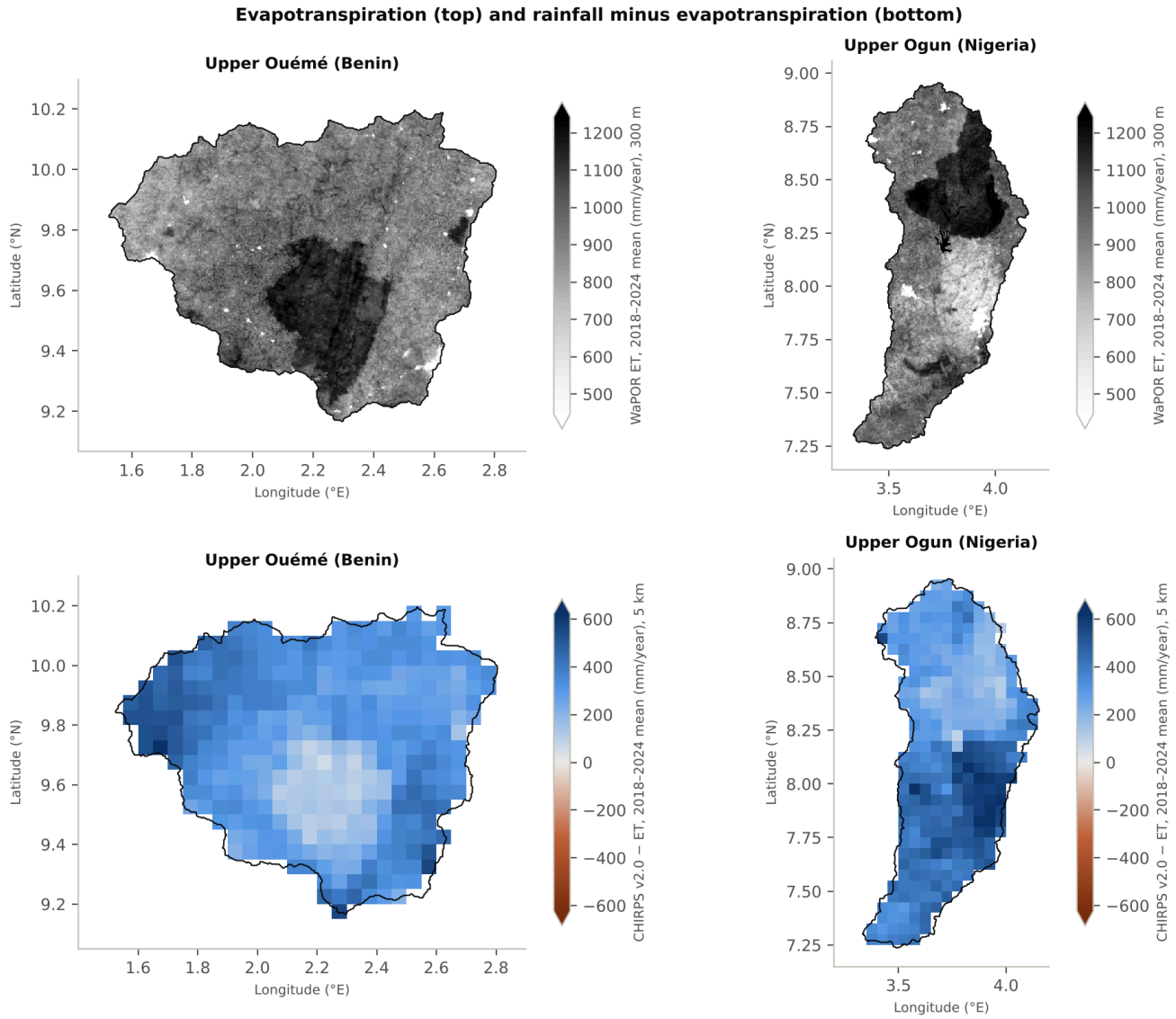


Figure 4. WaPOR v3 evapotranspiration (top) and CHIRPS v2.0 rainfall minus evapotranspiration (bottom), 2018–2024 mean, mm per year, on the 5 km CHIRPS grid. The web version shows these maps year by year.

Table 2. Upper Ouémé water balance by rainfall product, 2018–2024 mean, mm per year, with r = 0.142. P rainfall, ET evapotranspiration, Q estimated river flow (Eq. 1), ε residual (Eq. 2).

Rainfall product	P	ET	$P - ET$	Q	ε	ε / P, %
CHIRPS v2.0	1,257	945	312	178	134	10.6
CHIRPS v3.0	1,298	945	353	184	169	13.0
TAMSAT v3.1	1,243	945	298	176	122	9.8

4.3 Dependence on the runoff ratio

Because river flow is estimated, the residual is only as reliable as the runoff ratio. Each 0.01 added to the ratio lowers the mean residual by about 13 mm (Figure 5). The balance closes at a ratio of 0.24–0.27, depending on the product. That is 1.7–1.9 times the ratio observed in 1971–2010, but inside the 0.14–0.28 found in single years in the Donga sub-basin (Kamagaté et al., 2007). Across that range the residual runs from 171 to −50 mm per year.

The ratio used here divides gauged flow by gauged rainfall, but is then applied to satellite rainfall. Recomputing it from each product's own 1988–2010 rainfall gives residuals of 121–179 mm per year: −1 to +10 mm different from those in Table 2, larger for both CHIRPS versions and slightly smaller for TAMSAT. Mixing the two kinds of rainfall therefore has little effect on the result.

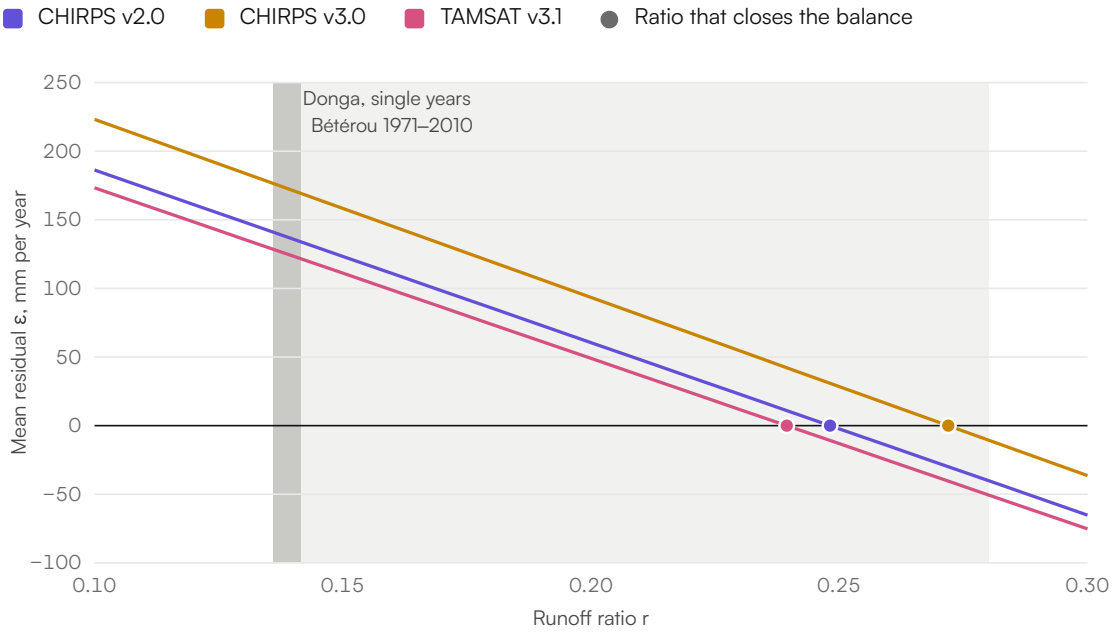


Figure 5. Mean 2018–2024 residual of the Upper Ouémé against the runoff ratio used to estimate river flow, for each rainfall product. Dark band: ratios observed at Bétérou in 1971–1987 and 1988–2010. Light band: single-year ratios in the Donga sub-basin, 0.14–0.28 (Kamagaté et al., 2007).

4.4 Storage cannot explain the residual

Summed over 2018–2024, the residual is 937 mm (CHIRPS v2.0). Storing it would need a water-table rise of 62–469 m for specific yields of 1.5% to 0.2% (Figure 6), in aquifers that lie mostly within the upper 100 m. Even a rise of 10 m would need a specific yield of 8.5–11.8%, several times the upper end of the range used. Whatever the residual is, it is not water accumulating in the aquifer.

62 m water-table rise needed to store the 2018–2024 residual

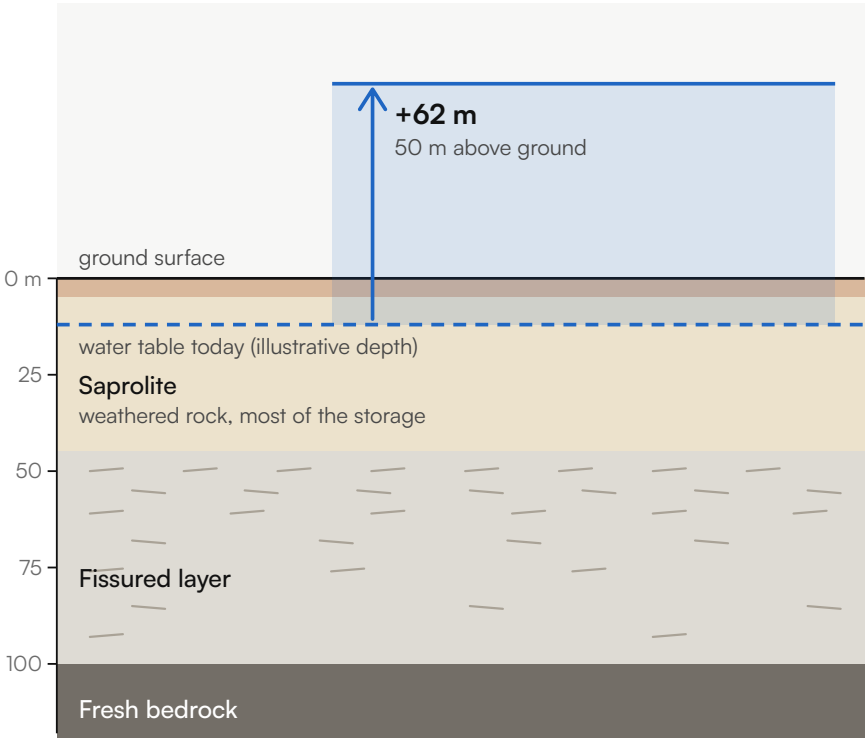


Figure 6. Water-table rise needed to store the cumulative 2018–2024 residual of the Upper Ouémé (Eq. 3). The profile is schematic, after Lachassagne et al. (2021), and does not represent observations; the present water table is drawn at an illustrative depth. Drawn for a specific yield of 1.5% and CHIRPS v2.0; the web version lets both be changed.

4.5 Comparison with published recharge

The residual is 9.8–13.0% of rainfall (Figure 7). That is inside the 5–24% reported for the Donga sub-basin (Kamagaté et al., 2007) and above the 4% from long-term hydrographs (Kotchoni et al., 2019). The uncertainty of the balance is therefore of the same size as the recharge it would be used to estimate.

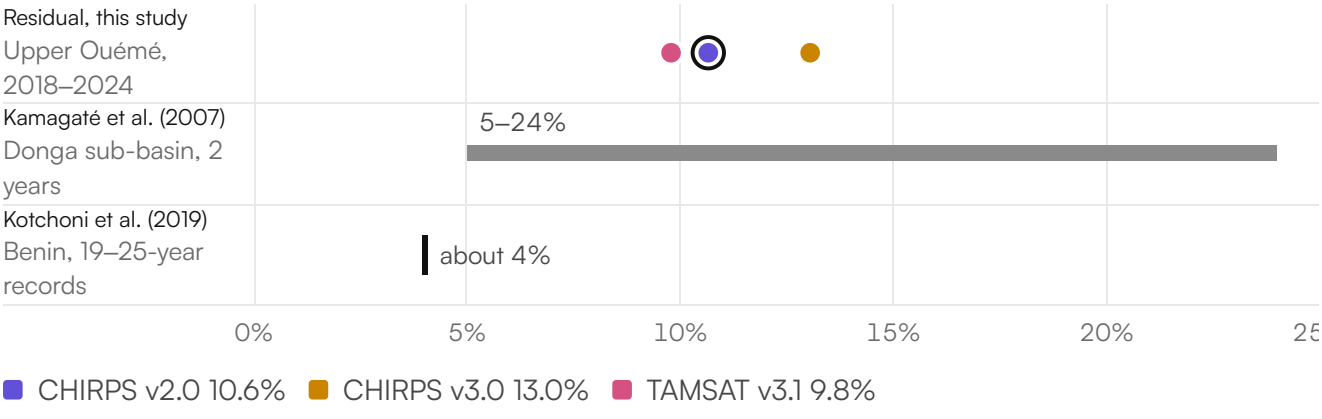


Figure 7. Upper Ouémé residual for each rainfall product, against recharge estimated from groundwater levels in Benin basement, as a share of annual rainfall. The published values come from other periods and, for Kotchoni et al. (2019), other sites.

4.6 Single-term changes that close the balance

Taken one term at a time, closing the 2018–2024 balance needs evapotranspiration 14.2% higher, rainfall 12.4% lower, or a runoff ratio of 0.25 instead of 0.142 (CHIRPS v2.0; Figure 8). Each change lies within reported uncertainty: 29–31% for WaPOR v2 evapotranspiration across Africa (Blatchford et al., 2020), 20–26% for rainfall-

product biases over Nigeria (Ogbu et al., 2020), and 0.14–0.28 for single-year runoff ratios in the Donga sub-basin (Kamagaté et al., 2007). The available data cannot say which term is in error, or whether several are.

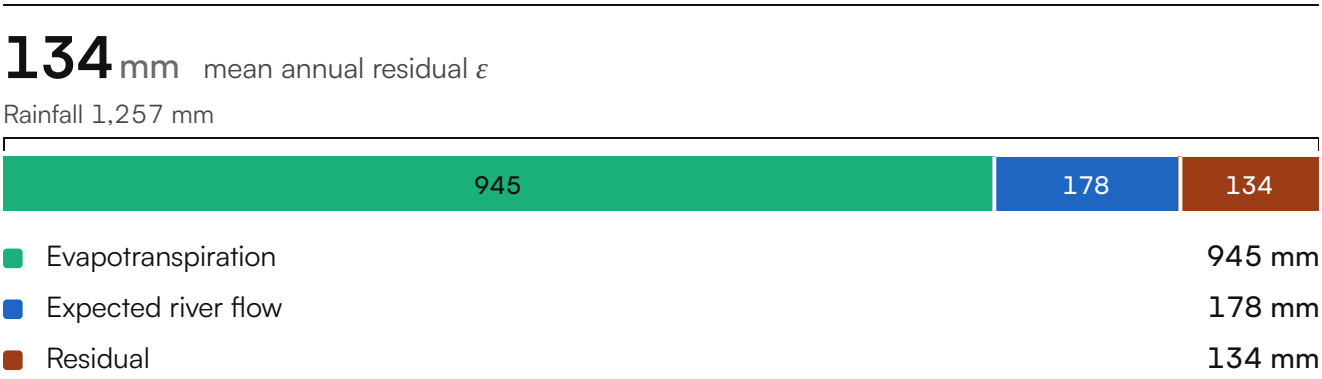


Figure 8. Upper Ouémé water balance, 2018–2024 mean, CHIRPS v2.0, with the runoff ratio of Table 2. The web version lets each term be changed.

4.7 Upper Ogun

Applied unchanged, the method gives 349–545 mm per year of rainfall minus evapotranspiration in the Upper Ogun (Table 3). No river-flow record was found, so the balance cannot be tested and no residual can be computed. The rainfall products differ more than in the Upper Ouémé: by 222 mm per year on average, against 105 mm. The difference between the catchments therefore lies mostly in the rainfall estimate; with the data available it cannot be attributed to climate, geology or land cover. These values are not recharge estimates.

Table 3. Upper Ogun water balance by rainfall product, 2018–2024 mean, mm per year. Without a river-flow record, Q and ε cannot be computed.

Rainfall product	P	ET	$P - ET$
CHIRPS v2.0	1,311	946	364
CHIRPS v3.0	1,492	946	545
TAMSAT v3.1	1,296	946	349

5 Discussion

5.1 What the residual contains

Over several years the residual equals the change in soil and groundwater storage, plus groundwater leaving the catchment underground, plus abstraction, plus the errors in rainfall, evapotranspiration and the estimated river flow. Storage is ruled out (Section 4.4). Underground outflow is assumed small in these shallow aquifers, and abstraction in the Donga sub-basin is a small share of recharge (Kamagaté et al., 2007); neither was measured here. What remains is error, and the largest single candidate is the river flow: it was not measured, and the ratio that closes the balance is within the range observed in single years nearby. A higher runoff ratio in 2018–2024 than in 1988–2010 is plausible, because the period was wetter and runoff ratios tend to rise in wet years, but it has not been shown. Evapotranspiration is the second candidate. WaPOR v3 has not been evaluated in these catchments, and the evaluation of v2 found errors larger than the change needed here.

5.2 Net and gross recharge

Satellite evapotranspiration includes water drawn from the water table by deep roots and wet valley bottoms. Even a closed satellite balance would therefore measure net recharge, while the water-table fluctuation method measures gross recharge. In the Donga sub-basin, recharge of 5–24% of rainfall coexisted with a weak groundwater contribution to river flow (Kamagaté et al., 2007), which is consistent with much of the recharge returning to the atmosphere. A satellite balance and a well-based estimate would not be expected to agree even if both were exact.

5.3 Relation to other estimates

That the residual is of the same magnitude as published recharge is not evidence that it is recharge. Across Africa, long-term recharge scales with long-term rainfall (MacDonald et al., 2021), and so does every error that is a fixed share of rainfall, including the error of a transferred runoff ratio. A residual of 9.8–13.0% of rainfall is what an unmeasured flow term of that size would produce whether or not any of it recharged the aquifer.

5.4 Implications

For basement catchments of this size, the uncertainty of a satellite water balance is as large as recharge. Such a balance can show where rainfall exceeds evapotranspiration and how products disagree, but it cannot estimate recharge without measured river flow for the same years and an independent check from wells. The order of the work matters: the balance should first be closed against measured flow, and only then interpreted.

6 Limitations and future work

Table 4 summarises how much each source of uncertainty that could be quantified changes the mean residual. The ranges are the results of alternative assumptions, not statistical confidence intervals.

Table 4. Sources of uncertainty in the Upper Ouémé residual, 2018–2024 mean, mm per year. Each row changes one assumption and keeps the others as in Table 2.

Source	Alternatives tested	Residual, mm per year
Rainfall product (Table 2)	CHIRPS v2.0, CHIRPS v3.0, TAMSAT v3.1, with r = 0.142	122–169
Runoff ratio, observed	r = 0.136 (1971–1987) instead of 0.142	129–177
Runoff ratio, rainfall basis	Each product's own 1988–2010 rainfall instead of gauge rainfall	121–179
Runoff ratio, single years	0.14–0.28, Donga sub-basin (Kamagaté et al., 2007)	−50 to 171
Evapotranspiration	WaPOR v3 ± 10%	27–264
Catchment area	10,070–10,143 km² instead of 10,102 km²	121–170

- No measured river flow for 2018–2024 was available. The runoff ratio of 1988–2010 is assumed to apply, and changes in land cover or abstraction since 2010 were not assessed.
- No river-flow data were found for the Upper Ogun, so its balance cannot be tested, and the two catchments can be compared only in rainfall and evapotranspiration.
- WaPOR v3 has not been evaluated against ground measurements in these catchments; the cited accuracy is for v2.
- The rainfall products are not independent, and no gauge record for 2018–2024 was available to measure their error.
- The specific-yield range is a regional value from the literature, not a measurement in these catchments.

- 2018–2024 was wetter than 1991–2020, and seven years are too few to describe long-term behaviour.
- The published recharge values come from other periods and, for Kotchoni et al. (2019), other sites.
- The hydrogeological description of the Upper Ogun is assumed from the regional setting.

Three data sets would resolve the residual. None has been analysed here.

1. River discharge at Bétérou for 2018–2024, which would replace the assumed runoff ratio and show whether the balance closes.
2. Groundwater levels from the AMMA-CATCH observatory, about 20 piezometers and 28 wells monitored in the Upper Ouémé since 1999 (Galle et al., 2018). The analysis code is written and tested on synthetic data only (`scripts/04_validate_wtf.py`); access requires registration.
3. Ground measurements of evapotranspiration. Eddy-covariance stations ran in northern Benin in 2008–2010, one inside this catchment (Mamadou et al., 2016), before the WaPOR v3 record begins.

7 Conclusion

RQ1. With the runoff ratio observed in 1988–2010, a satellite water balance does not close at Bétérou in 2018–2024: 122–169 mm per year are unaccounted for, in every year and with every product. Because river flow was not measured in those years, this shows that the balance cannot be closed with the data available, not that the satellite products are wrong.

RQ2. The residual is not stored groundwater. It is error in one or more terms, most plausibly the estimated river flow or the evapotranspiration, and it is as large as published recharge. A runoff ratio within the range observed in single years nearby removes it.

RQ3. In the Upper Ogun the method gives rainfall minus evapotranspiration only. The products disagree there by about twice as much as in the Upper Ouémé, and nothing can be said about recharge.

The next step is measured discharge at Bétérou for the satellite years, followed by the AMMA-CATCH well records. Until then, satellite water balances in these catchments describe the surplus of rainfall over evapotranspiration, not recharge.

Data and code availability

All scripts, the analysis outputs, a dataset inventory (research/data_inventory.csv) and a register of assumptions (research/assumptions.md) are in the repository github.com/oloyededavid/Oueme-Ogun-Groundwater-Recharge. Every number in this report is computed from outputs/summary.json and outputs/annual_balance.csv, which scripts 01–03 produce from openly available data. The Bétérou flows are taken from Kodja et al. (2020); the underlying daily records are not public.

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